RH56DFTP-E2 SOP

Inspire Robots' 4th-gen dexterous hands include two models: RH56DFTP and RH56E2.
Both dexterous hands share identical registers but differ in appearance and mechanical structure, so hardware inspections must be performed separately for each model.
This manual contains two chapters: **software operation** and **hardware structure**. Please distinguish between them as needed.
As a general rule, prioritize software troubleshooting first to identify fault types, then proceed with hardware inspection for follow-up after-sales handling.

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I. Software-Related Issues

1. Basic Operation Guide for Upper Computer Software

1.1 Basic Communication

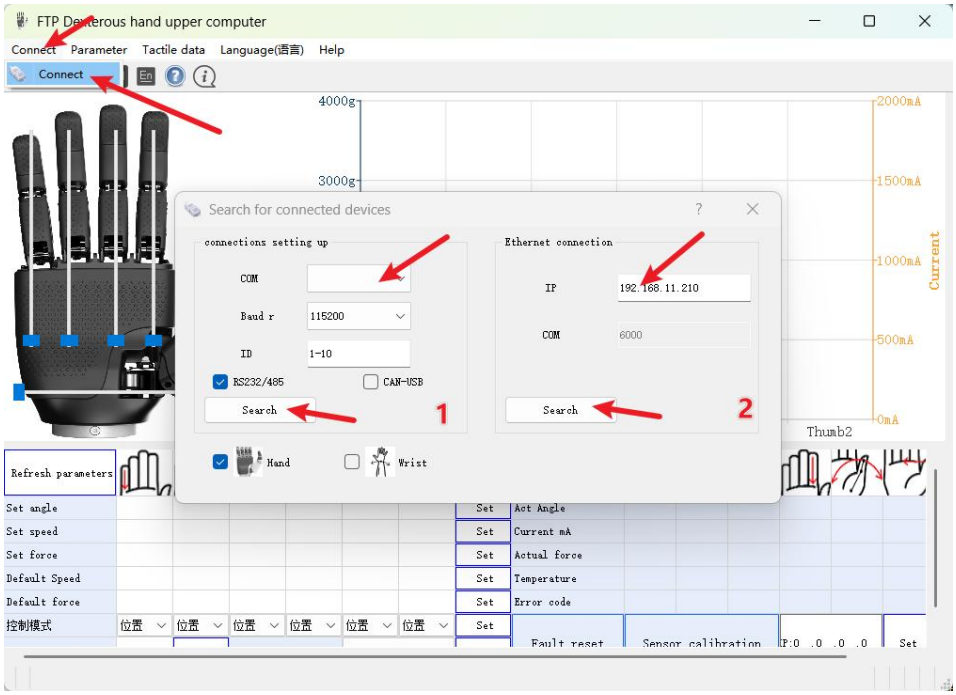
Power on the dexterous hand and connect it to a PC.

The dexterous hand supports multiple communication protocols: RS485/Modbus TCP and CAN/Modbus TCP.

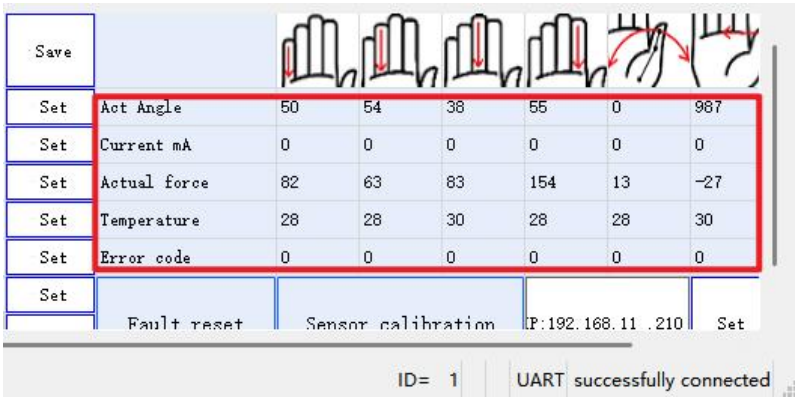
RS485/CAN Serial Communication (Mark 1 in the screenshot): Install the corresponding serial driver first. If the PC fails to detect the serial port after connection, install the CH340/CH341 driver (installation steps refer to the Driver Installation section).

Modbus TCP Ethernet Communication (Mark 2 in the screenshot): Set the PC's IP address to match the hand's network segment.

After selecting the correct connection mode, click Search to connect to the dexterous hand.



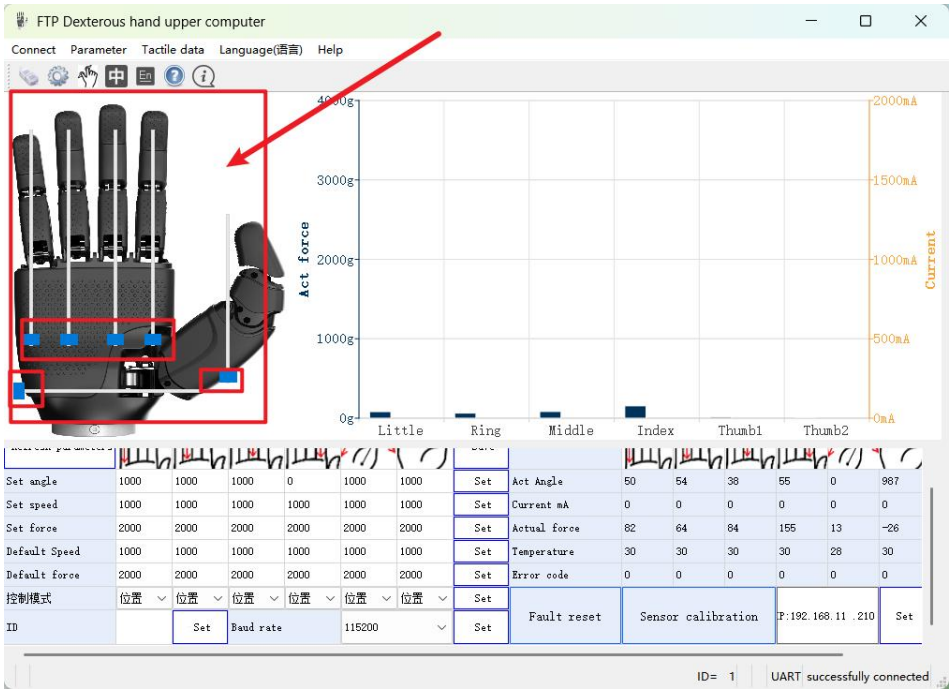
Upon successful connection, real-time data values will refresh in the lower-right panel.



1.2 Core Function Overview

Finger Control Sliders (Left Hand Diagram)

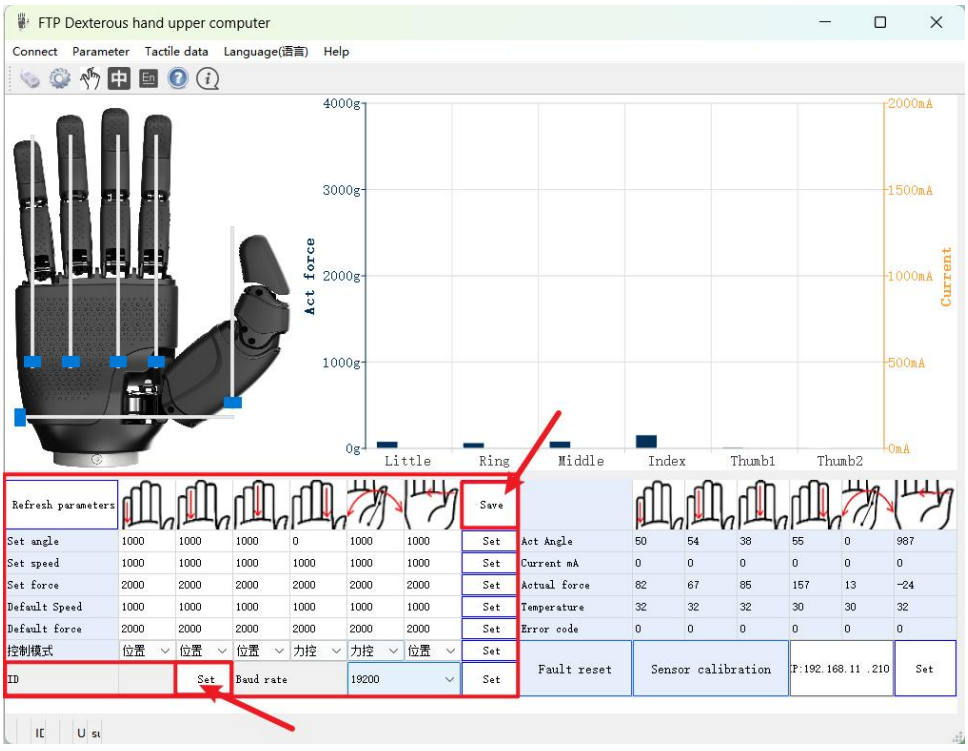
Drag the sliders corresponding to each finger degree of freedom to manually control finger movement.



Parameter Settings (Lower-Left Area)

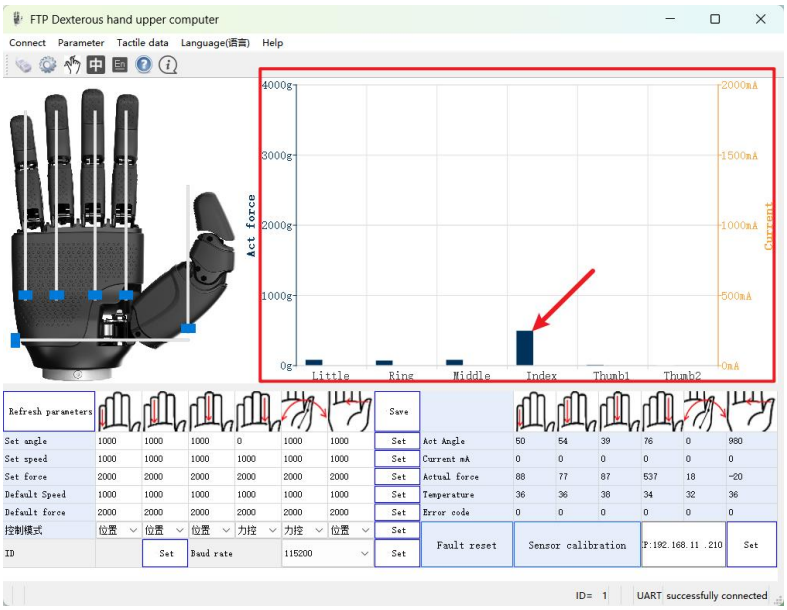
Adjust all dexterous hand parameters within the specified range; valid ranges and control modes are detailed in the official user manual.

To quickly modify the device ID: enter the target ID, click Set, then click the central Save button to apply changes.



Force & Current Histogram (Right Panel)

Real-time histograms display current gripping force and actuator operating current for each finger.



Real-Time Status Data (Lower-Right Area)

Live readouts reflect the hand's operating status.

Fault Code Column: Displays error codes when actuators malfunction, enabling rapid fault diagnosis.

Error code	0	0	0	0	0	0

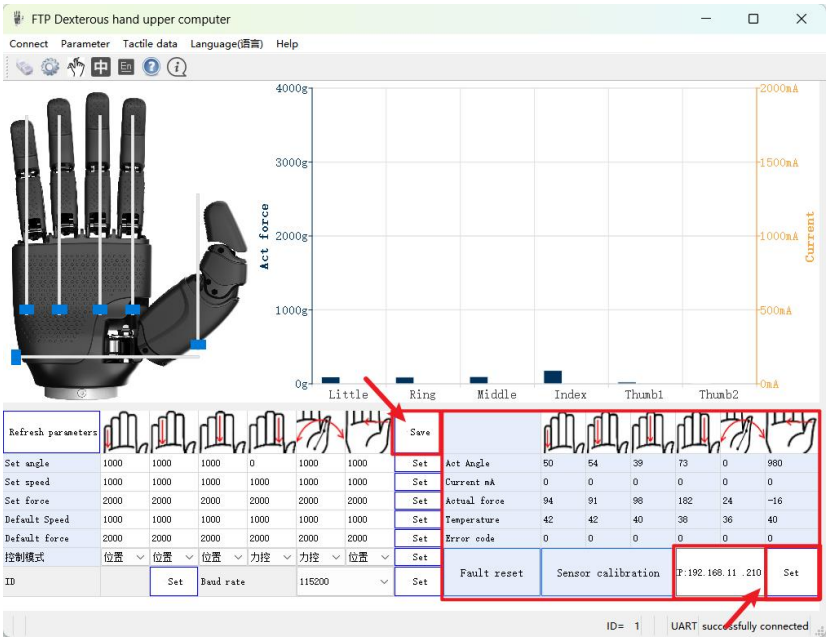
Functional Buttons:

Fault Reset: Clear actuator fault alarms

Sensor Calibration: Correct finger offset drift



IP Configuration: Input the target IP, click Set, then Save to update the hand's Ethernet IP address.

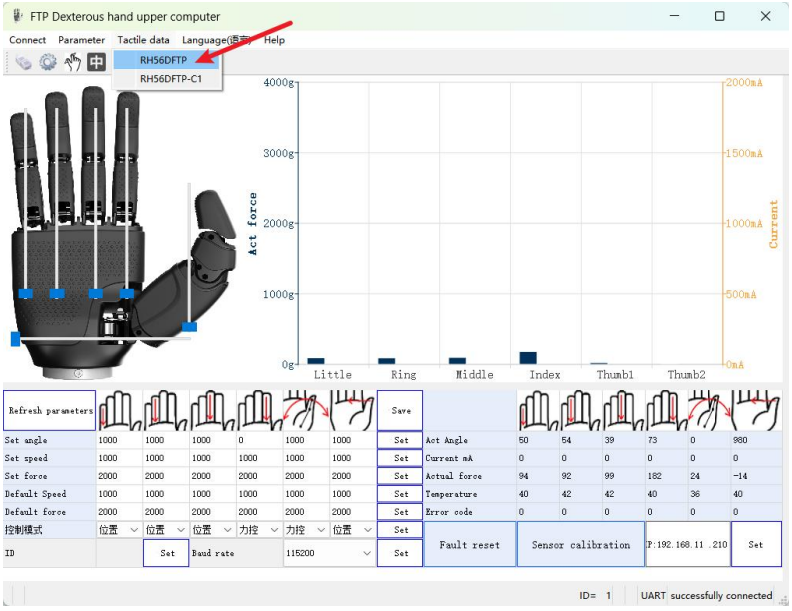


Tactile Data (Modbus TCP Only)

Navigate to the Tactile Data tab and select the matching model:

Resistive tactile version: RH56DFTP

Capacitive tactile version: RH56DFTP-C1



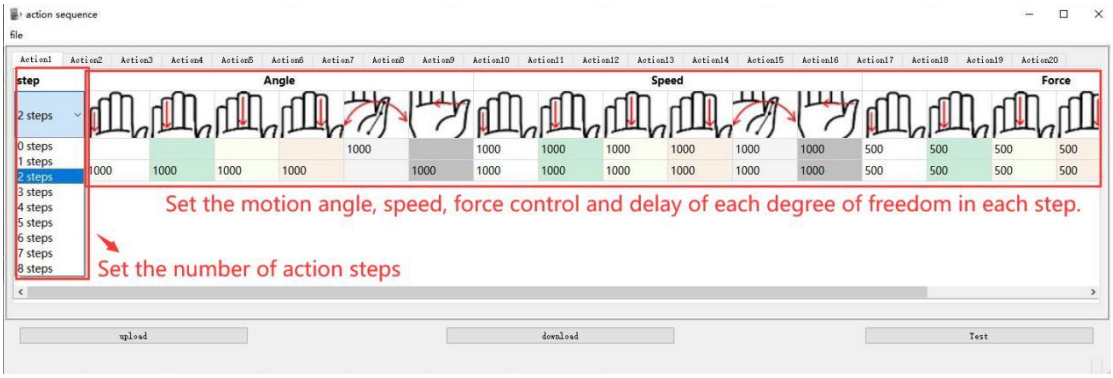
The tactile interface displays colored heatmaps and numerical readouts when pressure is applied to finger surfaces.



1.3 Motion Sequence Library

The software supports importing custom motion sequences, which can be saved locally on the dexterous hand.





The software can set the data of the 1st to 40th set of action sequences in the dexterous hand. Each set of action sequence can have up to 8 steps to decompose the action. The finger angle, movement speed, strength and waiting time of each step of the decomposed action can be set separately. Step number: the number of steps in the action sequence, which can be set to 0-8;

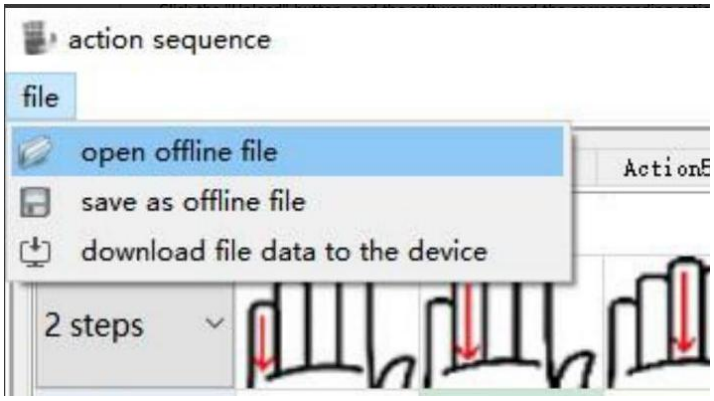
Finger angle: set the finger target angle of a step in the action sequence. The angle input range is 0-1000, 0 corresponds to the maximum bending of the finger, 1000 corresponds to the full opening of the finger, and the space represents the finger does not move. Motion speed: set the finger motion speed of a step in the action sequence, 1-1000. Force setting: set the finger force of a step in the action sequence, 1-1000. Delay: set the finger movement speed of a step in the action sequence. After setting the data of an action sequence, click the "Download" button to download the

data to Dexter. Click the "Test" button to control the dexterous hand to trigger the corresponding action

sequence for movement. Click the "Upload" button, and the software will read the corresponding action sequence of

the dexterous hand and display it in the table.

1.4 Action sequence file operation



"Open offline file" can open the data of an action sequence data file and display it in the action sequence data area of the software.

"Download the file data to the device", you can download the data of an action sequence data file in batches to dexterous hands.

"Save as offline file" can save the whole action sequence data displayed and edited by the software to the specified file.

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2. Finger Offset / Fingers Unable to Move

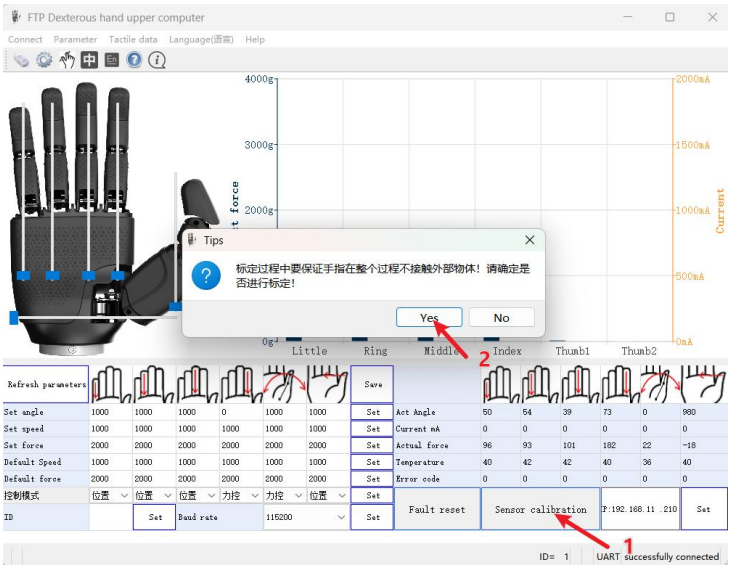
Individual finger lockup or drift is typically caused by collision, heavy impact, or uncontrolled robot arm movement during transit or operation.

If only positional offset occurs, perform sensor calibration via the upper computer first:

Establish stable communication with the hand; fully extend all fingers (ignore unresponsive fingers temporarily).



Click the Sensor Calibration button in the lower-right panel. A pop-up prompt will appear“保证手指在整个过程不接触外部物体”-ensure no external objects contact the fingers during calibration. Then click Yes. Wait ~10 seconds for calibration to complete, then test finger movement. If the fault persists, proceed to hardware inspection.



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3. Communication Failures: IP Connection Failed / 485 Communication Failure / CAN Communication Failure

The 4th-gen hands support dual communication protocols. If one protocol fails, test the alternative to isolate software vs. hardware faults.

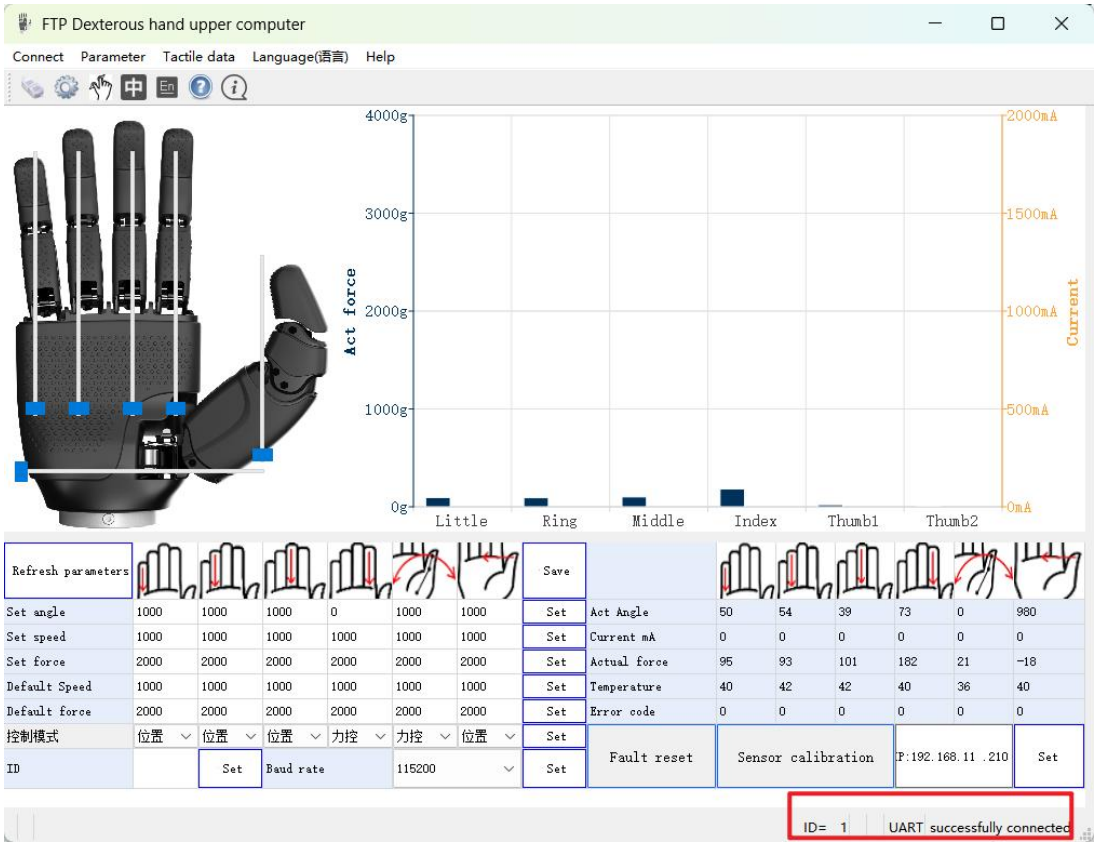
Common Root Causes

Mismatched IP addresses after mainboard replacement, incorrect manual IP edits, or damaged Modbus TCP modules.

Troubleshooting Steps

Use a USB-to-485 / USB-to-CAN adapter (included with official debug cables or purchased separately) to connect via serial communication.

Successful serial connection will display the device ID and UART Connection Successful in the lower-right corner.



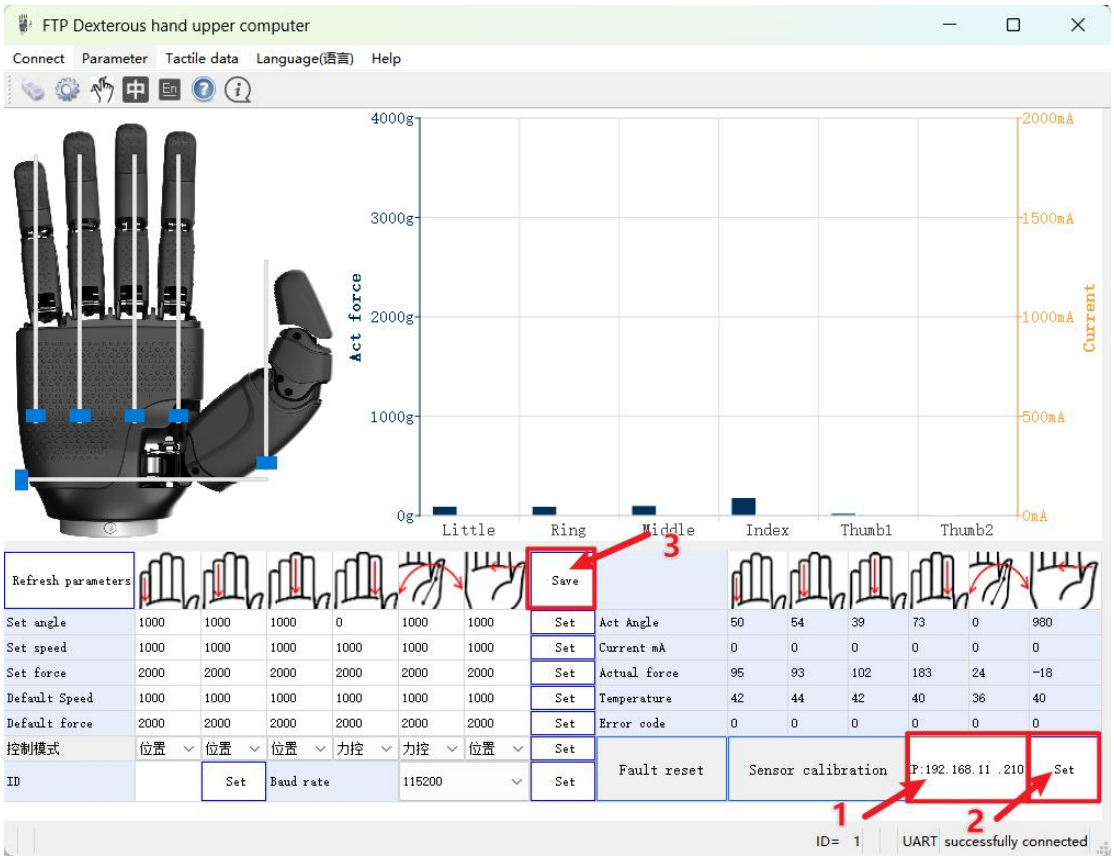
Confirm IP mismatch as the root fault: The factory default IP is 192.168.11.210.

If the IP mismatch your device, please modify the IP via serial connection with these steps:

Enter the target IP address in the IP input box

Click Set to apply temporary changes

Click Save to write the new IP to the hand's memory



Reconnect via Ethernet to verify the new IP works.

If connection still fails, disable VPNs or network monitoring software on the PC.

If all software troubleshooting fails, the fault originates from hardware; proceed to the hardware inspection chapter.

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4. Actuator Fault Alarm (Red LED on Back of Hand)

Normal operating status: Green steady LED on the hand’s rear panel.

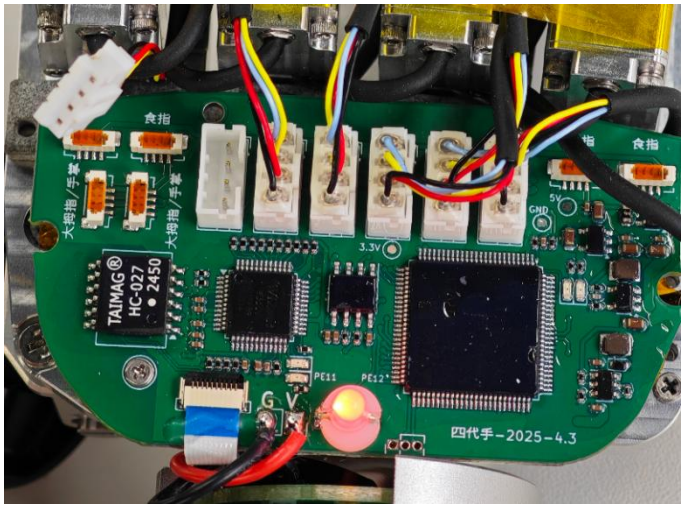
Red LED triggers when actuators throw persistent errors or communication faults.

Temporary recoverable faults auto-reset; if the same error occurs 3 times consecutively, the LED locks red. Clear alarms via upper computer Fault Reset or power cycle the hand.

The real-time data panel displays the faulty actuator ID and error type: stall, overcurrent, or overheating.

Non-destructive faults can be cleared remotely; adjust motion speed/force to prevent repeated damage to actuators.

Disconnected actuator wiring also triggers red LED and communication fault codes on the upper computer.



Specific error can be seen at the bottom of the software

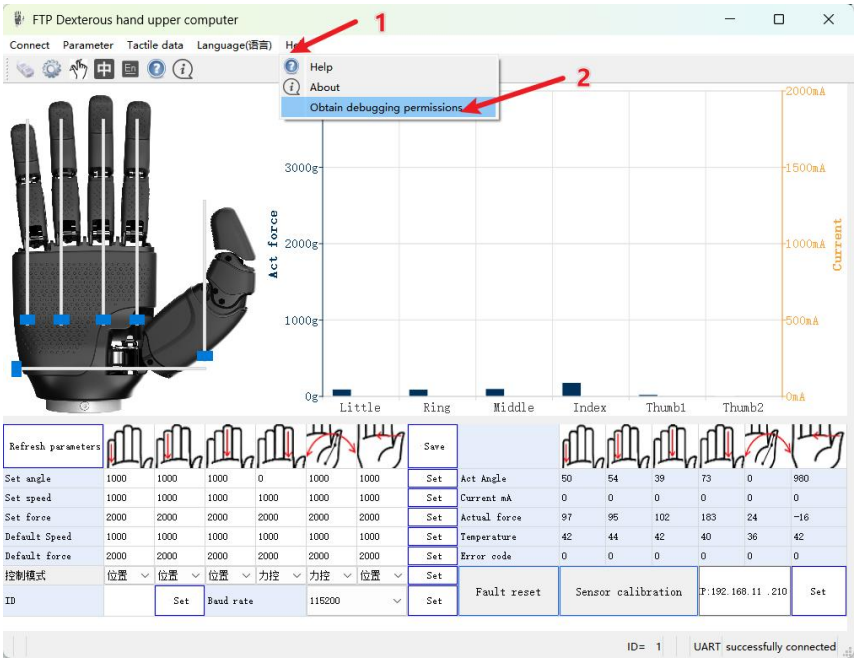


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5.Supplement: Internal Parameter Tuning Guide (Not Recommended for Unnecessary Use)

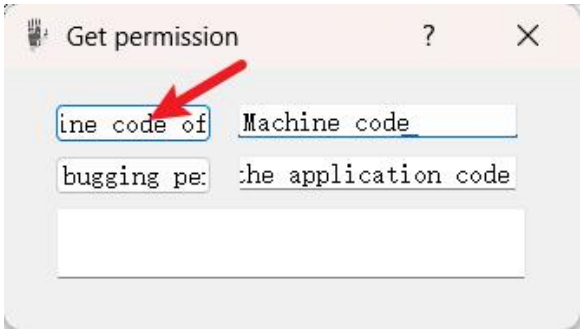
Use this feature only for severe finger offset or abnormal hand behavior after mainboard replacement.

Connect the hand via any communication protocol, then navigate to Help > Obtain Debug Permission.



In the pop-up window:

Click Get Machine Code and copy the generated string



Open the debug code provider and copy the machine code to the provider, then click arrow 2

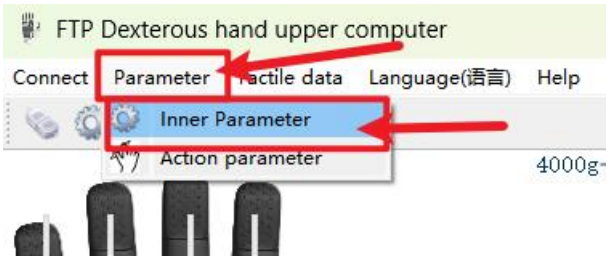


Then copy the debug code to the permission section and click arrow 2
If it was activated successfully, it would report “激活成功”.

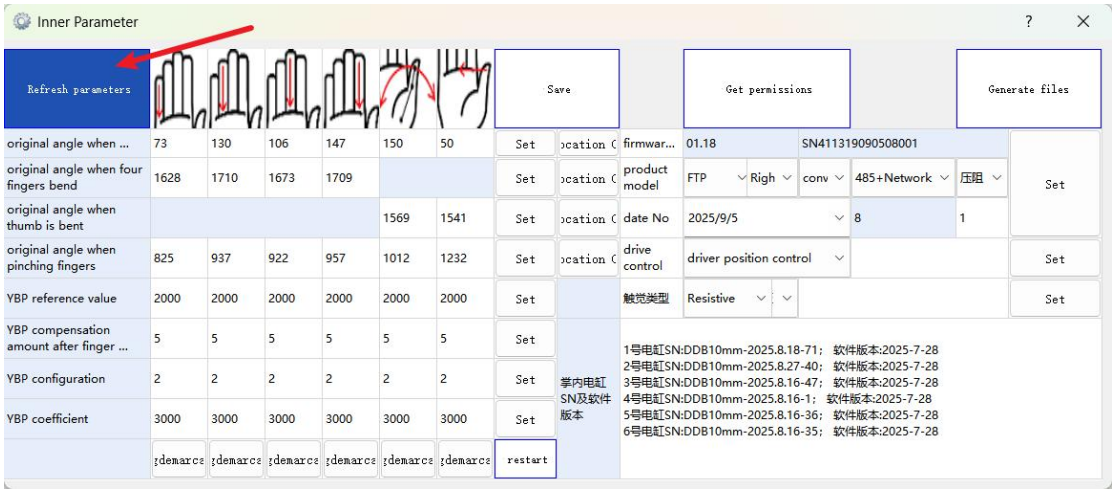


After that, please **restart the host computer software** to activate the debug permission

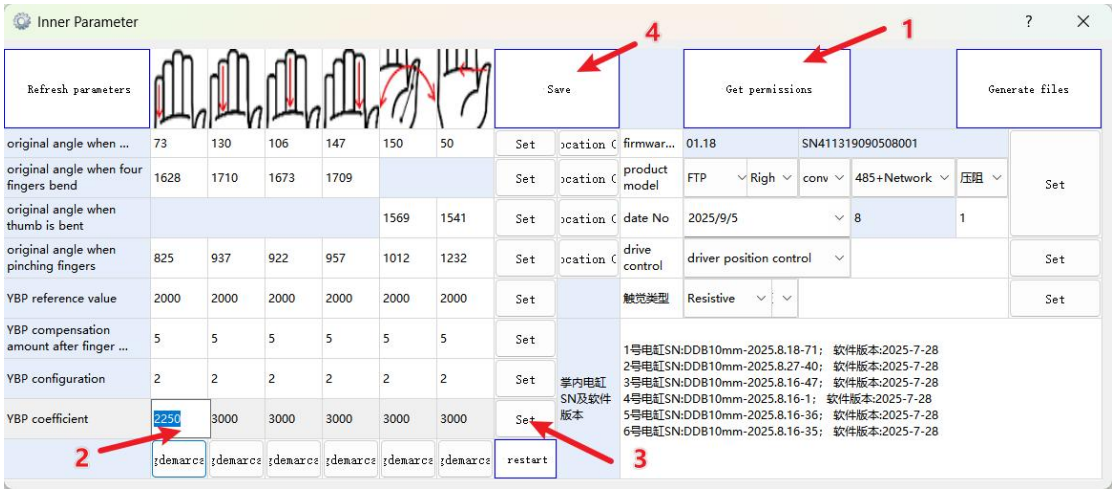
Inner parameter adjustments
Click “Parameter” and select “Inner parameter”.



Click refresh parameters to get the default parameters of the hand.



Click get permissions
Select the parameters to make adjustments,
Change the values, then click Set to apply the change.
Click Save to permanently store new parameters on the hand



After changing the value, please make a quick test to check if your problem had been solved.

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II. Hardware-Related Issues

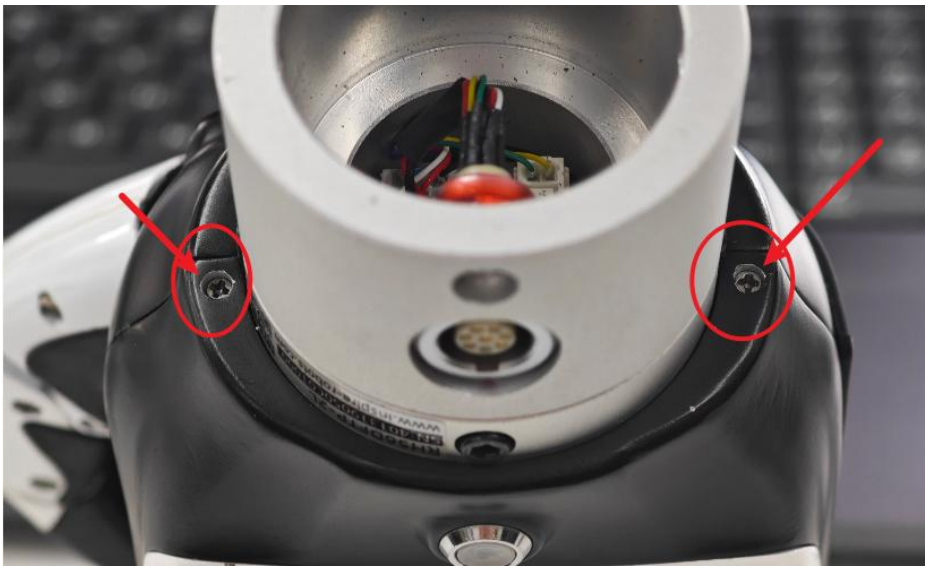
Disassembly of 4-Finger & Side Swing Actuators (RH56DFTP)

Required Tools: Cross-head screwdriver, hex key wrench

1.Connect the upper computer, fully extend all four fingers, and move the thumb side swing to ~50% travel.



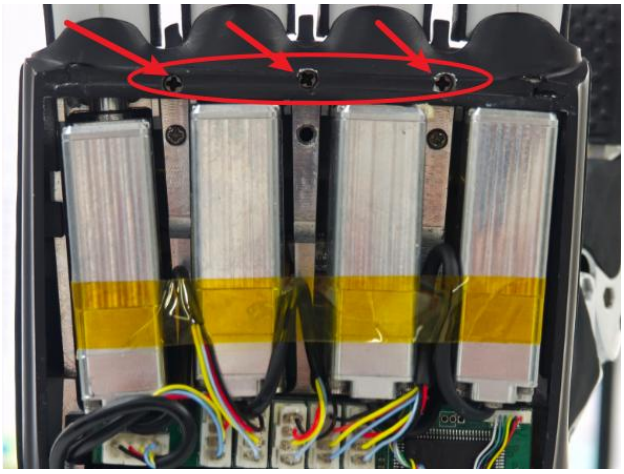
2.Remove the two fixing screws on the bottom sides of the palm base.



3.Remove the two rear panel fixing screws on the top of the hand’s back shell.



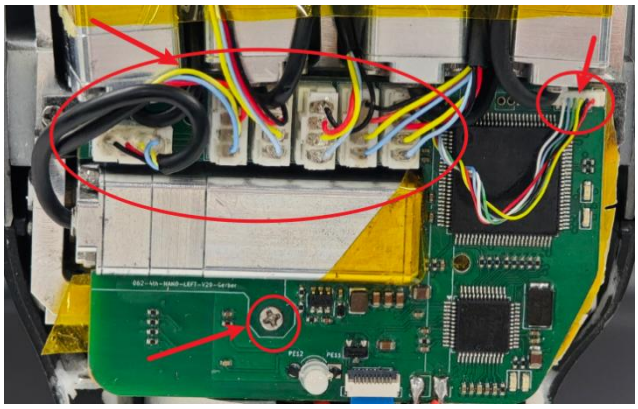
Remove the outer rear cover, then unscrew the three frame fixing screws beneath it.



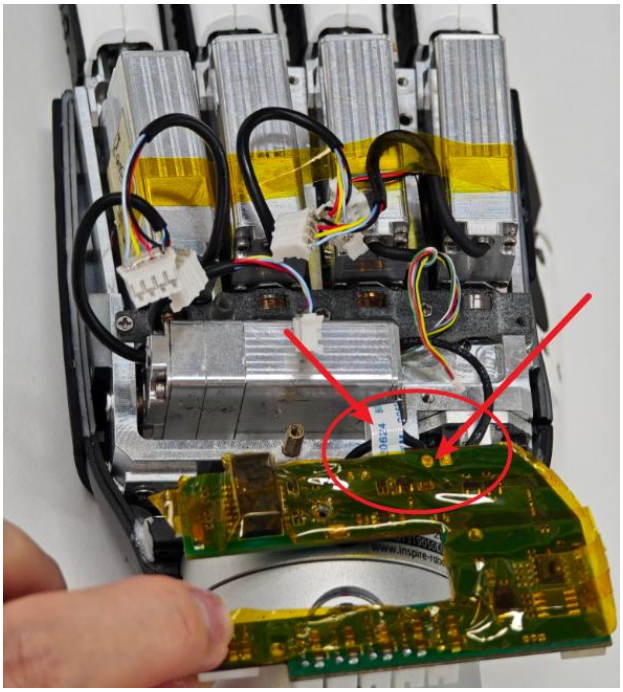
The full internal rear structure is exposed after frame removal.



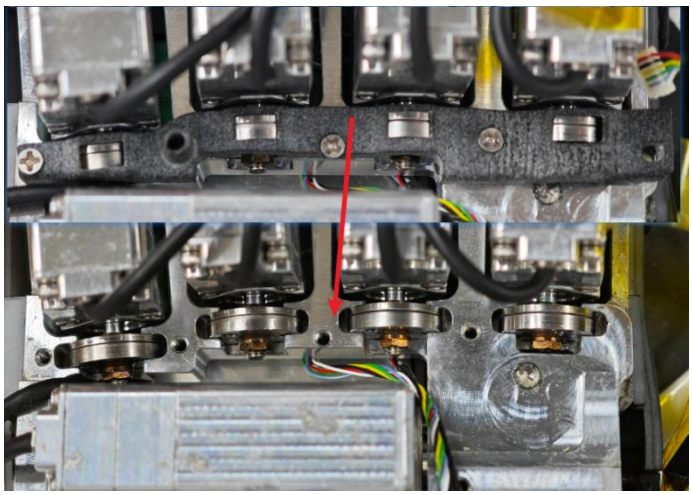
4.Unplug all actuator and tactile sensor wiring connectors, then remove the mainboard fixing screw.



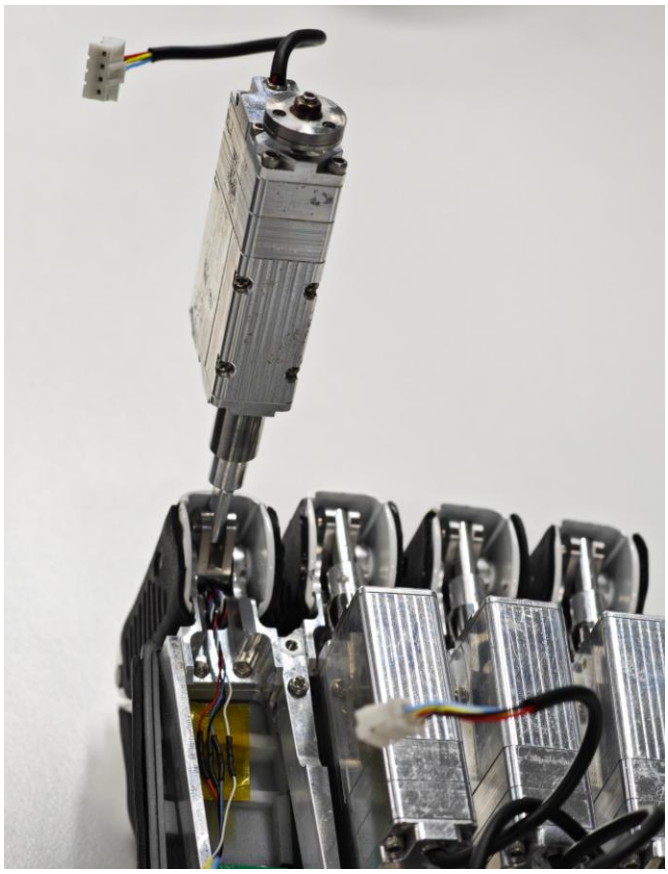
Carefully lift the mainboard to avoid tearing thin flat cables and power wires.



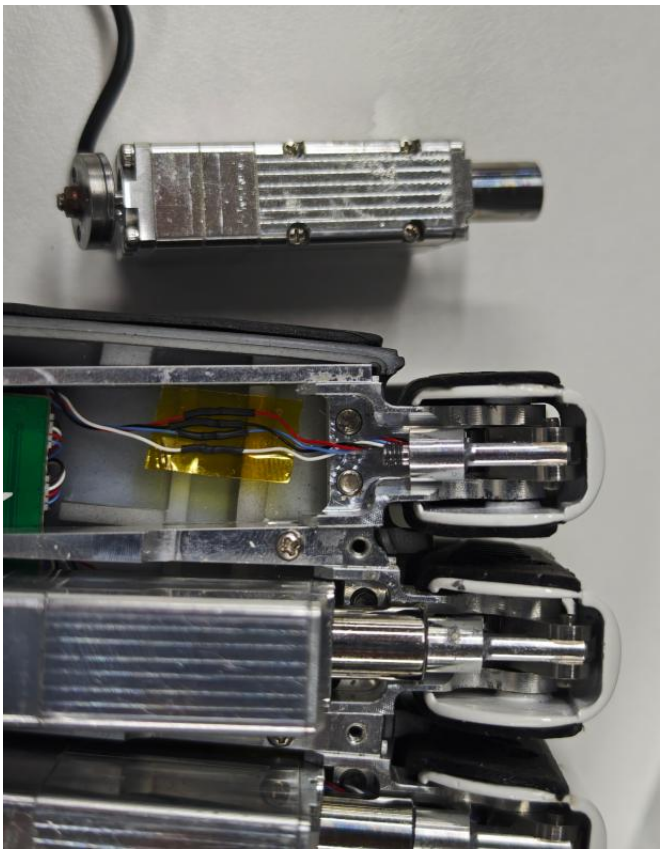
5.With the mainboard removed, access the actuator lock plate; unscrew the lock plate fixing screws.



The actuator can be removed after lock plate removal.



Rotate the actuator to detach it from the finger linkage for replacement.

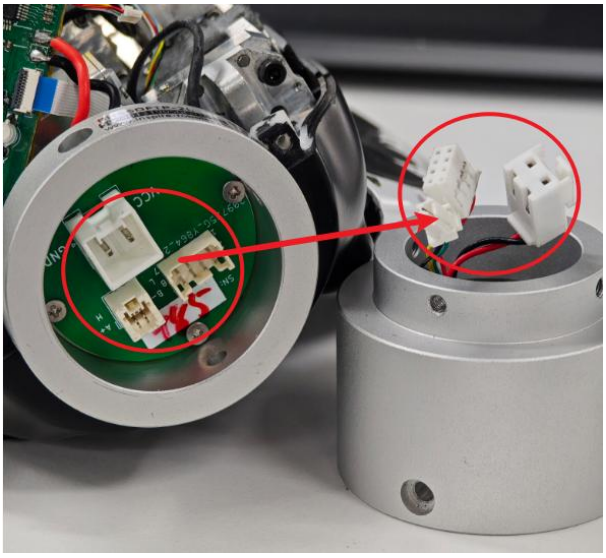


6.Side Swing Actuator Disassembly (Further Full Hand Disassembly)

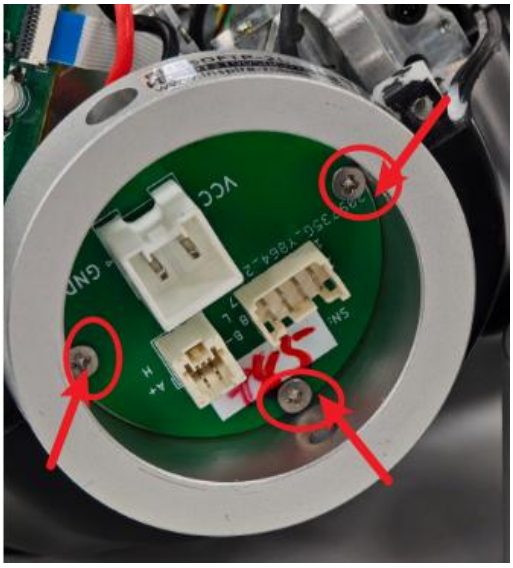
Unscrew the four wrist base fixing screws.



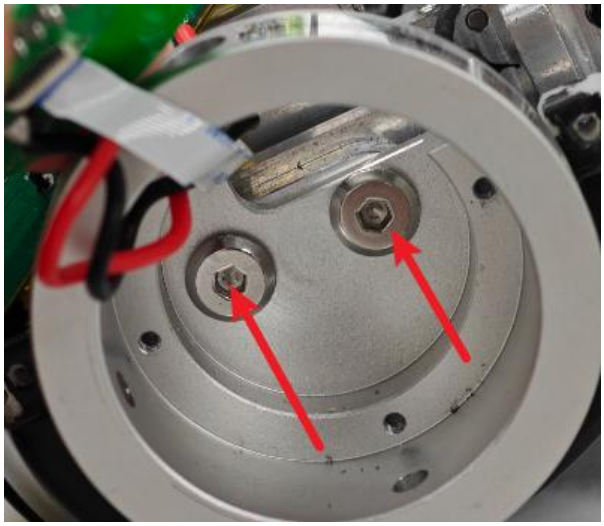
Unplug all power and signal connectors from the wrist adapter board.



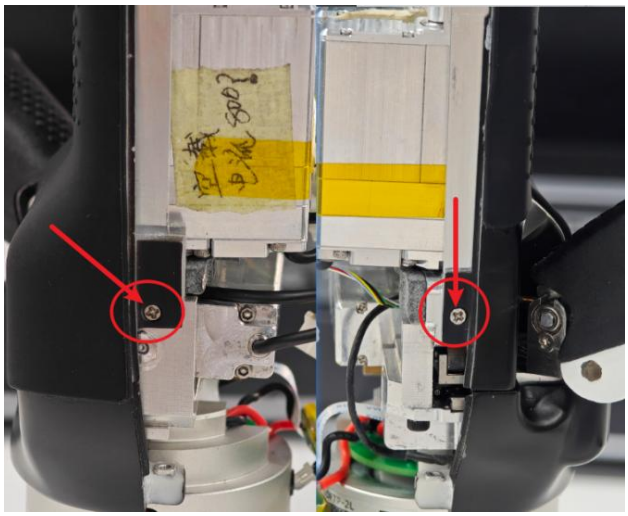
Remove the three adapter board fixing screws.



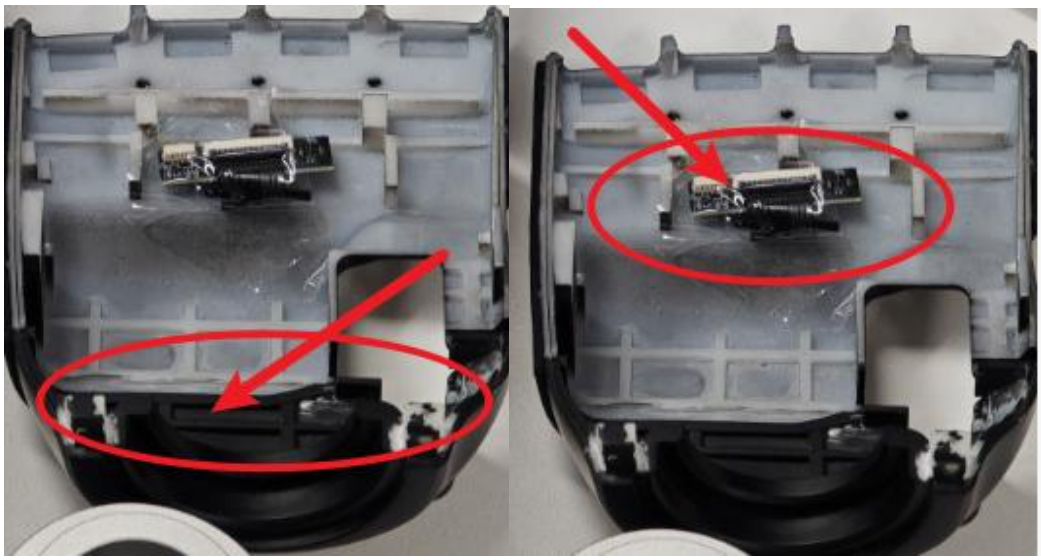
Detach the adapter board and remove the two internal base fixing screws underneath.



7.Unscrew the two front cover fixing screws on both sides of the palm.

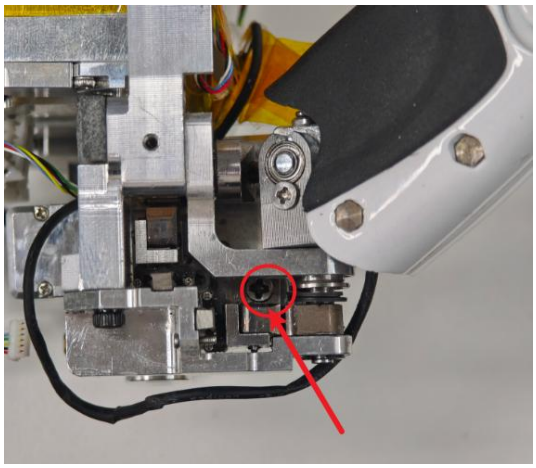


Slowly remove the front cover: the cover is bonded with flexible rubber gaskets (prone to tearing if forced) and carries a tactile sensor flat cable that must be unplugged first.

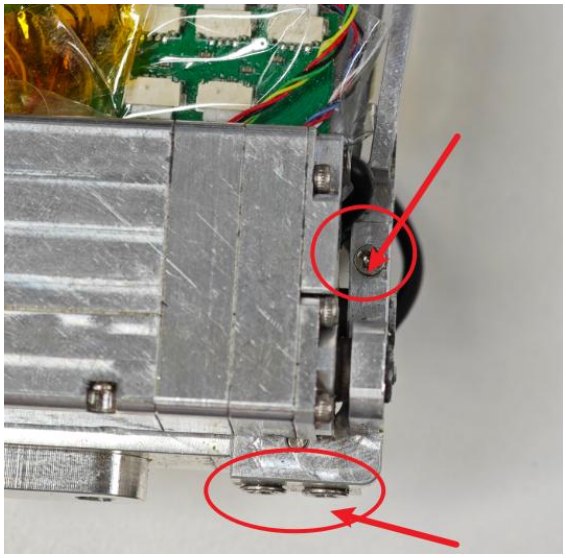


8.Side Swing Actuator Extraction

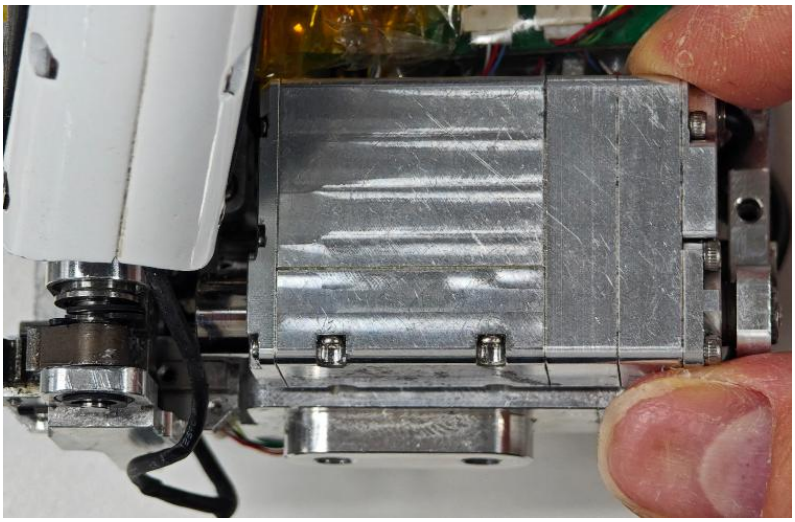
Unscrew the push rod linkage fixing screw first.



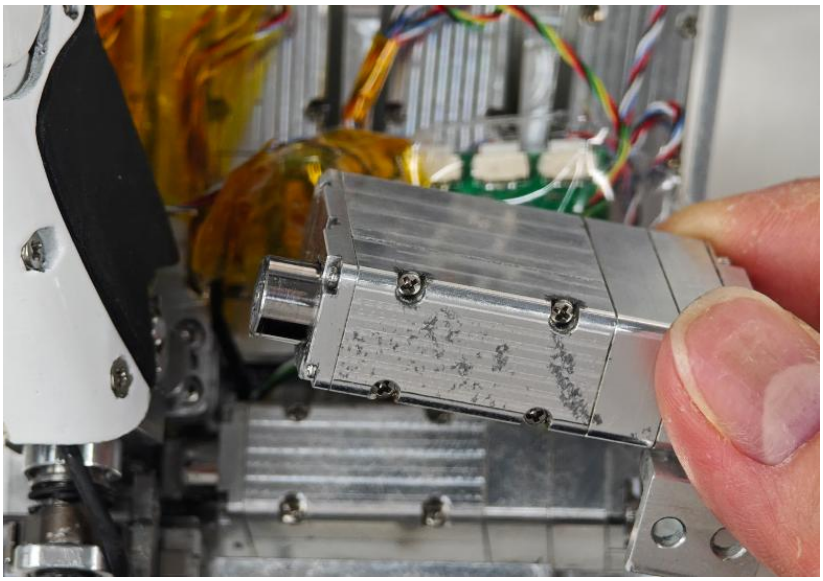
Remove the three lock plate screws on the actuator tail.



Grip both sides of the actuator tail, rock gently side-to-side, and pull upward to extract the side swing actuator.



Install the new actuator in reverse order of disassembly.



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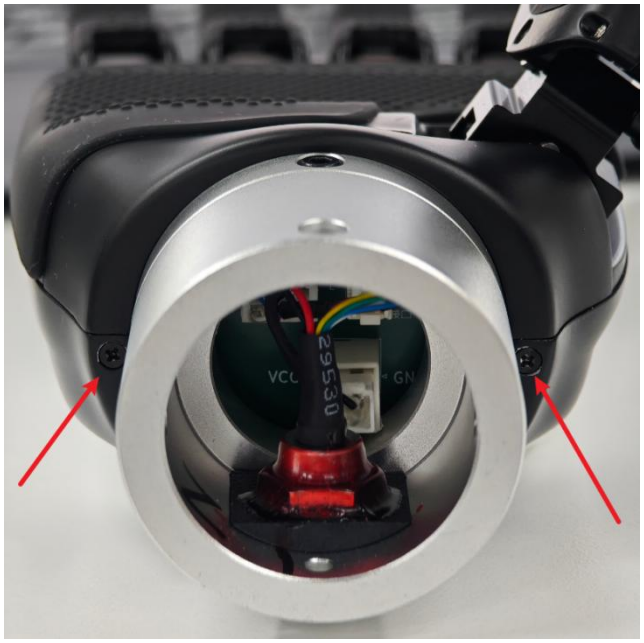
Disassembly of 4-Finger & Side Swing Actuators (RH56E2)

Required Tools: Cross-head screwdriver, hex key wrench

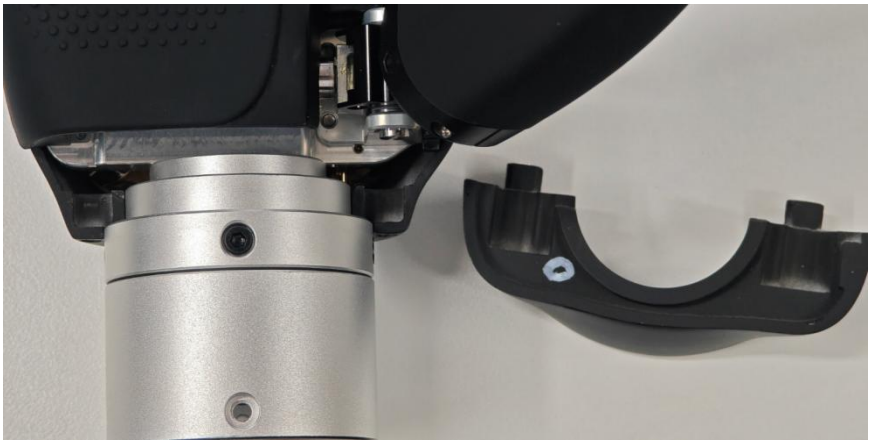
1.Connect the upper computer, fully extend all four fingers, and move the thumb side swing to ~50% travel.



2.Remove the two bottom palm base screws.



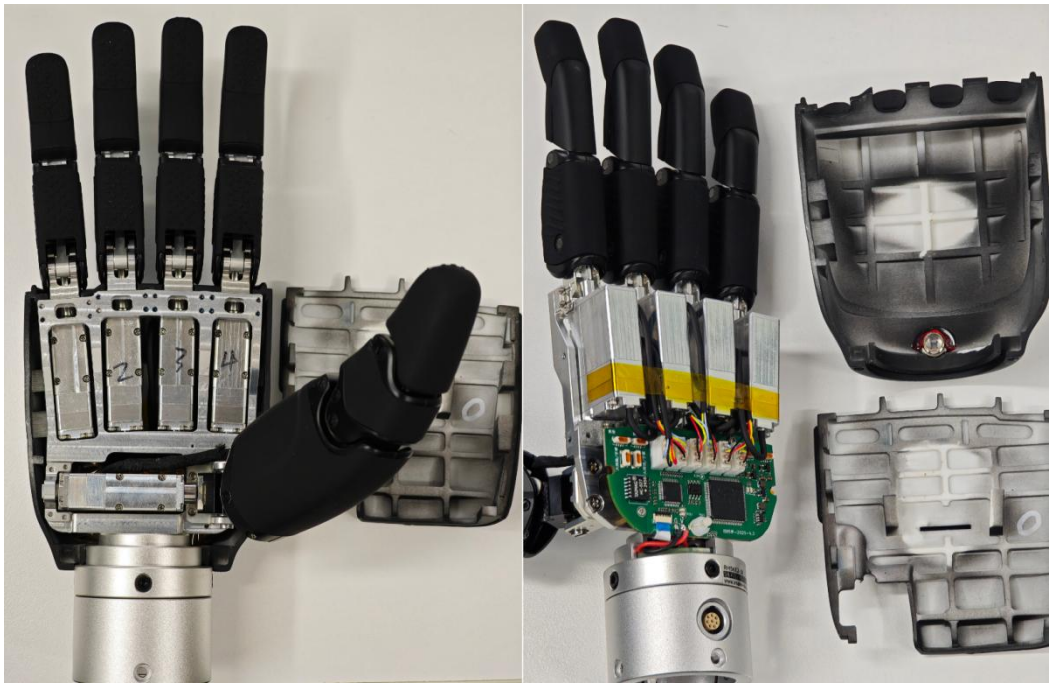
Detach the wrist bottom protective cover.



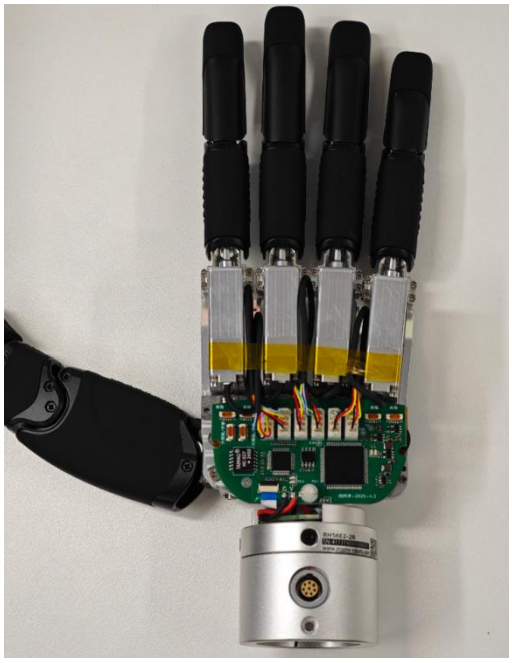
3.Unscrew the four side shell fixing screws (two on each side of the hand).



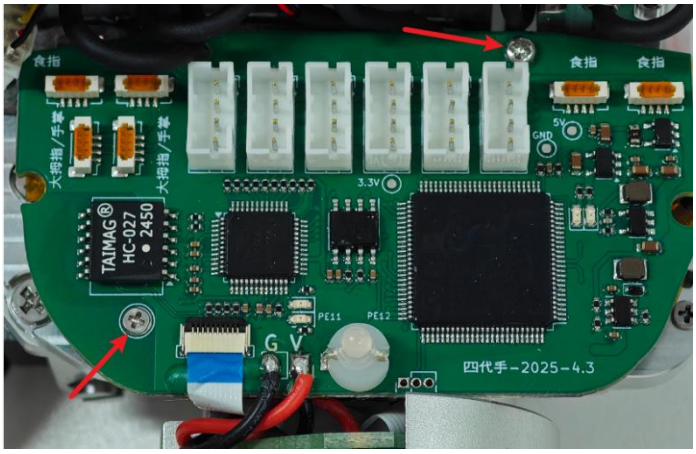
Separate the front and rear outer shells to expose the internal actuator assembly.



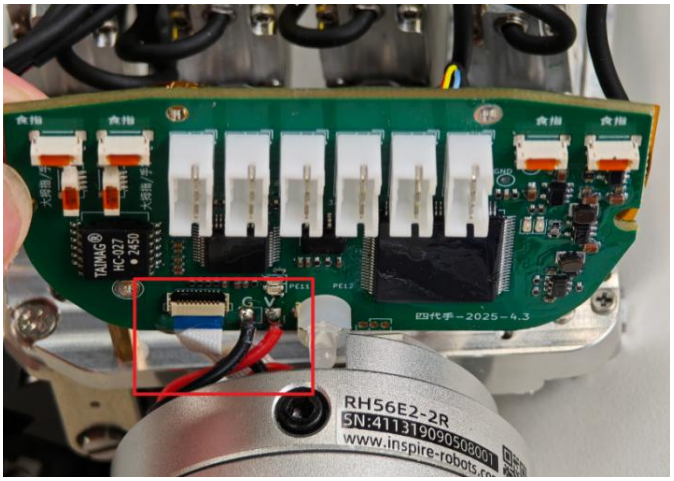
The complete display of the back condition is shown. As the sample does not have a tactile function, the current tactile module is in an empty state



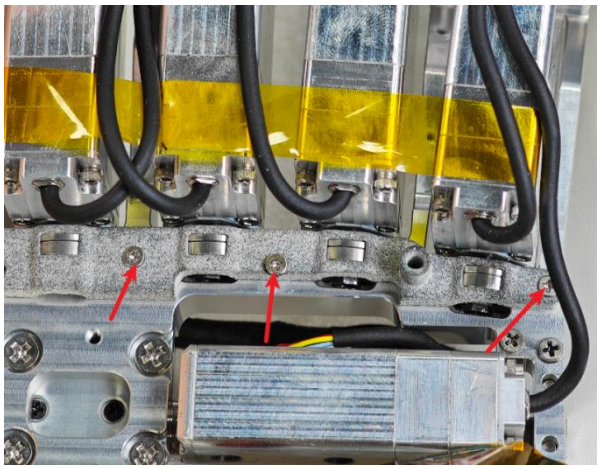
4.Unplug all actuator and tactile sensor wiring (label wires for easy reassembly), then remove the two mainboard fixing screws.



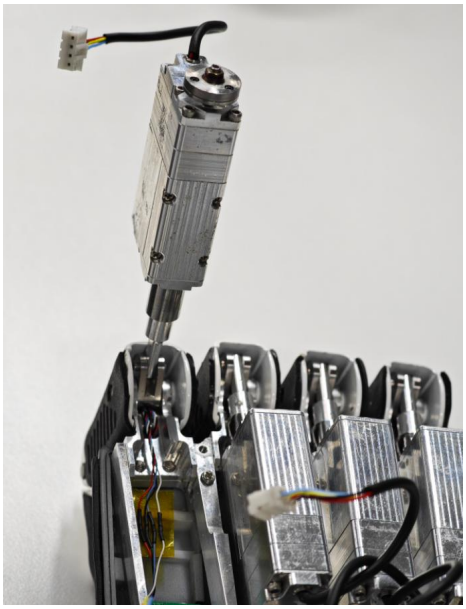
Lift the mainboard carefully to avoid damaging power and flat cables.



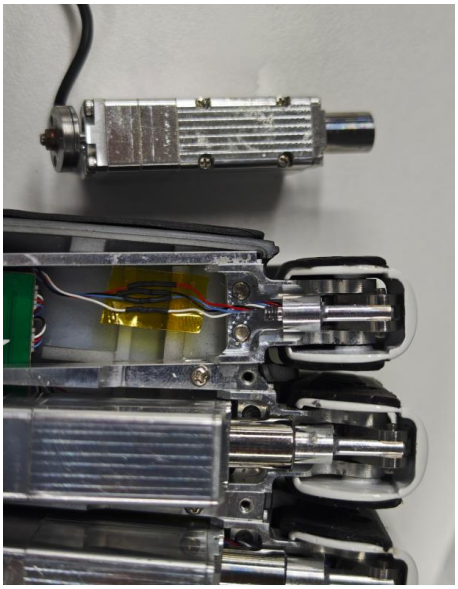
5.Unscrew the three actuator lock plate fixing screws below the mainboard.



You can take out the actuator after the lock is removed



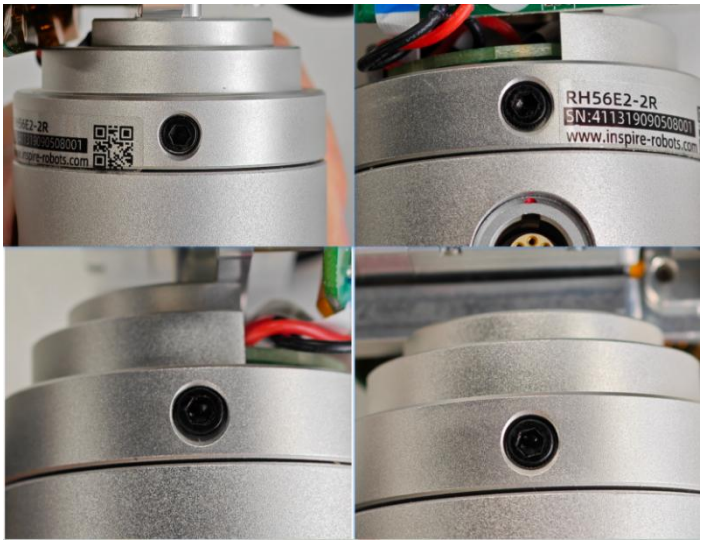
Extract the 4-finger actuator assembly and rotate to detach from finger linkages for replacement.



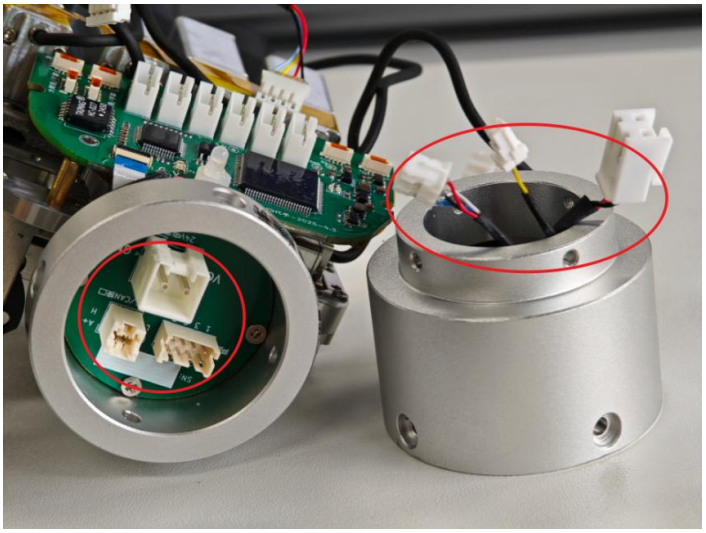
6.Side Swing Actuator Disassembly

Note: Wrist base screws may be factory-locked with thread locker; skip this step if full wrist disassembly is undesirable.

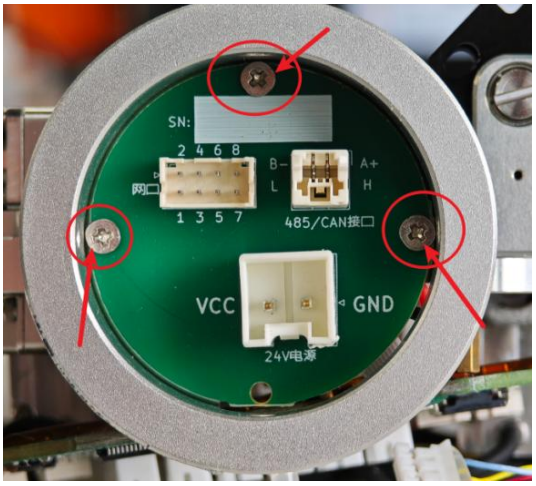
Unscrew the four wrist base fixing screws.



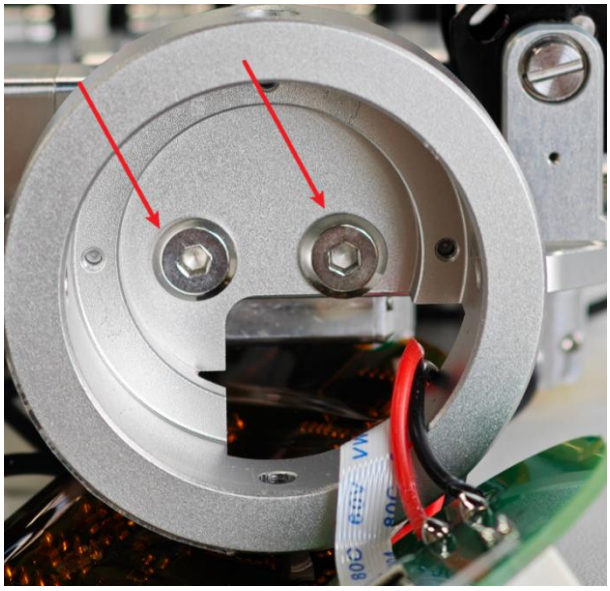
Unplug all wrist adapter board connectors.



Remove the three adapter board mount screws and detach the board.

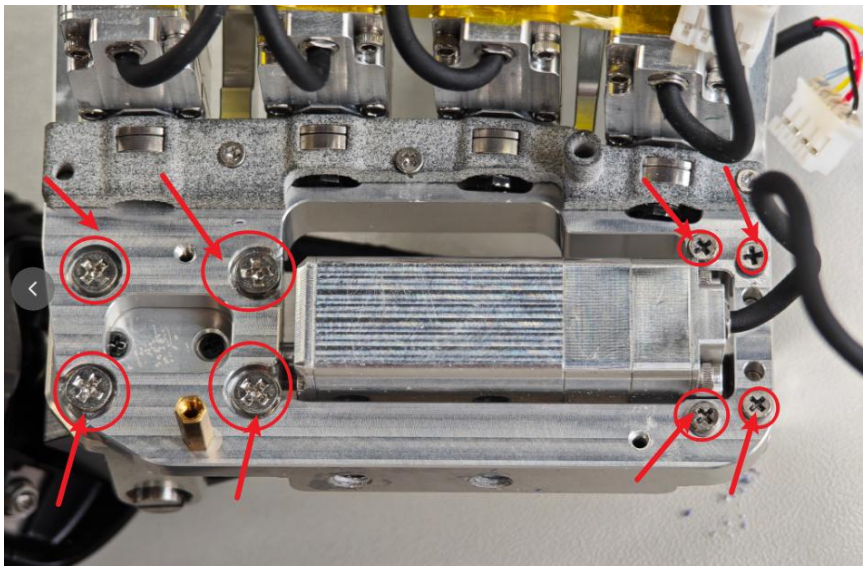


Then remove the two internal base fixing screws underneath.

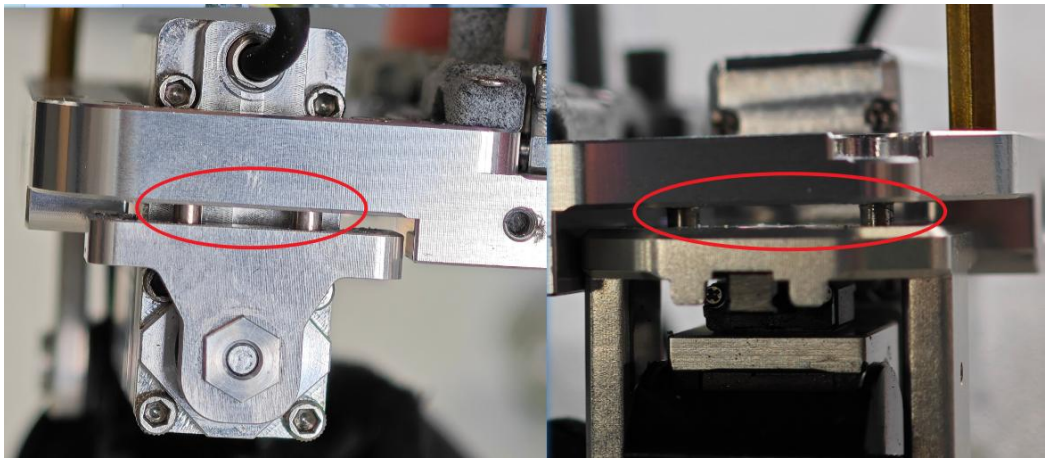


7.Side Swing Actuator Extraction

Unscrew all eight side swing actuator frame fixing screws.



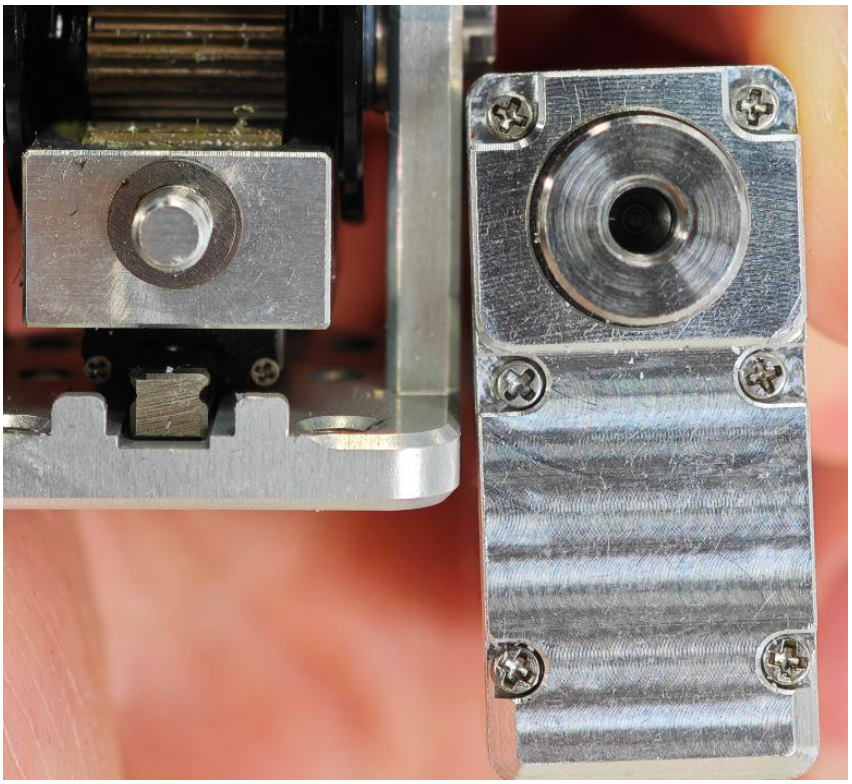
The actuator lock frame uses pin fasteners; rock the frame gently after screw removal to extract the full thumb-side swing subassembly.



Thumb and actuator can be removed



Rotate the actuator to separate it from the linkage for replacement.



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2. Mainboard Short Circuit (No LED Light on Back of Hand)

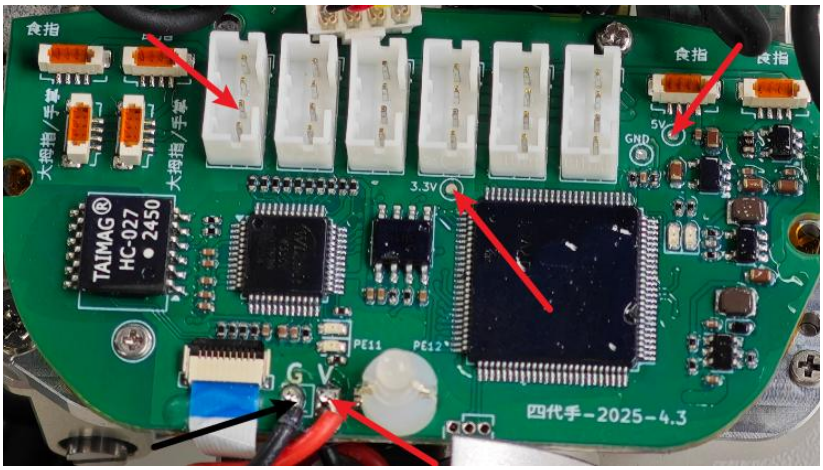
When powered by standard 24V supply and the rear LED remains off, two primary fault modes exist:

Case 1: Mainboard Short Circuit or Component Burnout

Visually inspect the mainboard for burnt components, discoloration, or burning odor.

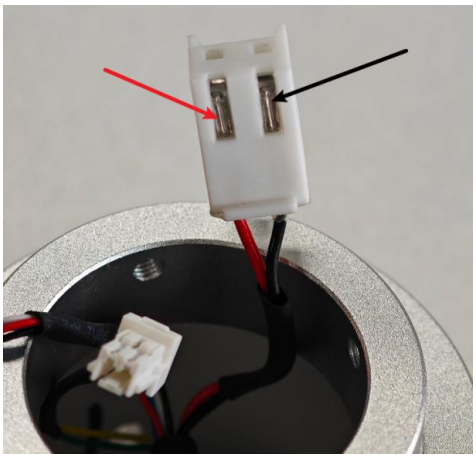
Power off the hand, unplug all actuator wiring from the mainboard, and use a multimeter to test continuity at the marked power test points. If a short circuit is detected, the mainboard is damaged.

After checking the mainboard, please inspect all disconnected actuators for internal short circuits, if short circuit happens please refer to Actuator Inspection section.



If no mainboard short is found, power on the hand and measure voltage at the marked test points. Zero voltage output indicates a faulty wrist adapter board; test continuity of all adapter board wiring terminals.

Measure the voltage at the wrist power input connector under power. If input voltage matches the power supply, the wrist adapter board is defective.



Case 2: Damaged Rear LED Indicator

If all power/communication functions operate normally but the LED does not light, only the LED component is damaged and can be replaced independently if hardware repair tools are available.

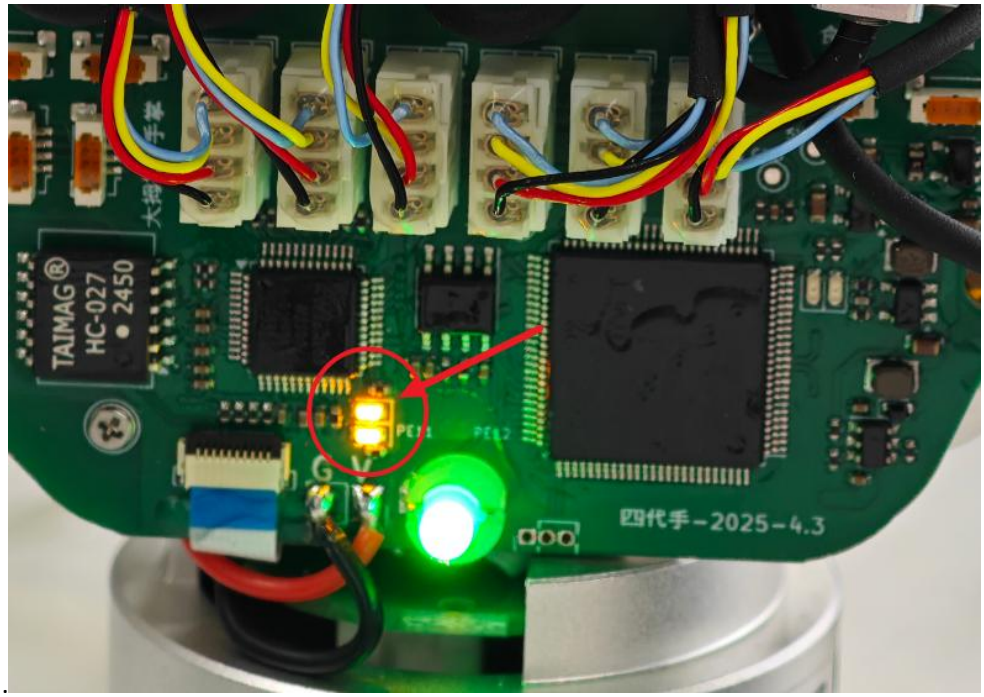
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3.Hardware Communication Failures: IP / 485 / CAN

Connection Errors

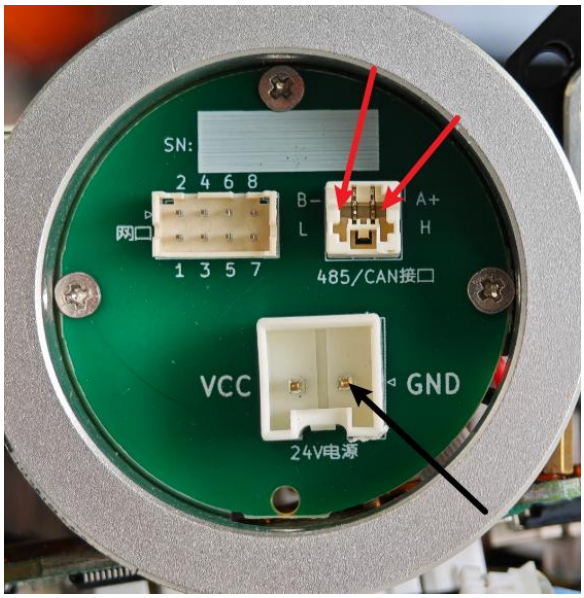
Ethernet (IP Communication) Troubleshooting

Two Ethernet status LEDs on the mainboard illuminate steadily when a valid TCP connection is established. If the LEDs remain off, the Ethernet module or wiring is damaged



485 / CAN Serial Communication Troubleshooting

Test continuity between the A+/B-/L/H terminals and GND on the wrist adapter board. If any communication pin is shorted to GND, replace the wrist adapter board.



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4. Actuator Inspection (Fingers Unable to Move)

1.Complete sensor calibration via upper computer to rule out positional offset faults.

2. Check actuator error status

If the finger still does not move after calibration, check the software for any actuator error (e.g., overcurrent, stall).

If an error is reported, try clearing it. Then power cycle the hand.

If the error persists, the actuator may be faulty.

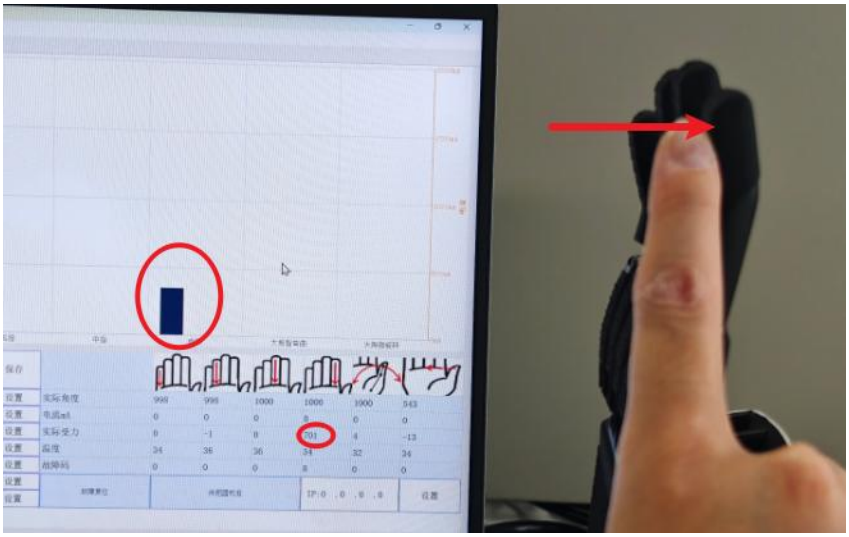
To confirm, remove the actuator from the finger (see step 4) and test the finger movement manually. If the finger moves freely, replace only the actuator.

3. Manual test – force reading

If no actuator error is reported, manually push/pull the finger while observing the force reading in the software.

If the force reading does not change, the actuator may be faulty.

If the force reading changes, proceed to step 4.



4. Disassemble and isolate

Open the back shell and disconnect the actuator from the finger.

Check the actuator:

Connect the actuator back to the main board. Use the software to control the corresponding actuator ID.

If the push rod extends and retracts normally, the actuator is good. Otherwise, replace it.

Check the finger structure:

With the actuator removed, manually move the finger mechanism.

If you feel obvious resistance or blocking, the finger structure is damaged and needs replacement.

If no resistance, take a known good actuator, install it into this finger, and control it via the software.

If the finger moves normally, replace only the actuator.

5.Actuator short circuit fault signature: Disconnecting one specific actuator restores mainboard LED and communication functions, while all other actuators operate normally.
Do not reconnect the shorted actuator to the mainboard to avoid secondary mainboard damage.
Use a multimeter to test for short/open circuits on the mainboard connector pins.
Test a spare actuator on the faulty port: Normal operation confirms only the original actuator requires replacement; failure indicates simultaneous mainboard damage.

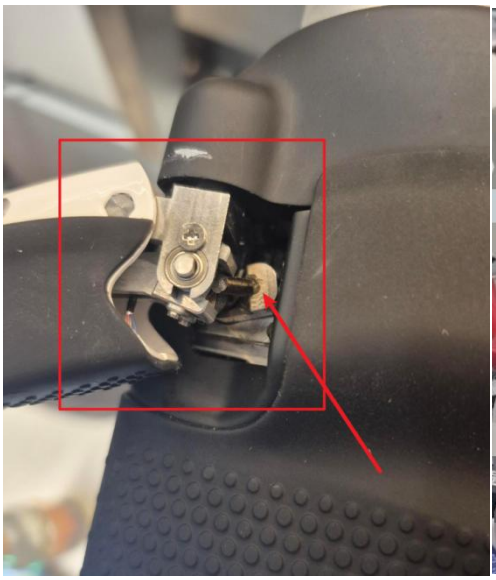
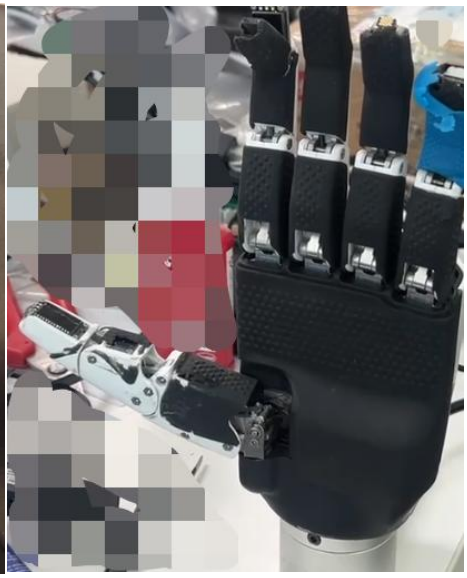
6.Reassembly
In rare cases, both the actuator and finger are good. Simply reassembling the actuator and finger may restore normal operation.

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5.Thumb-Related Faults (High Replacement Difficulty; Customer Self-Repair Not Advised)

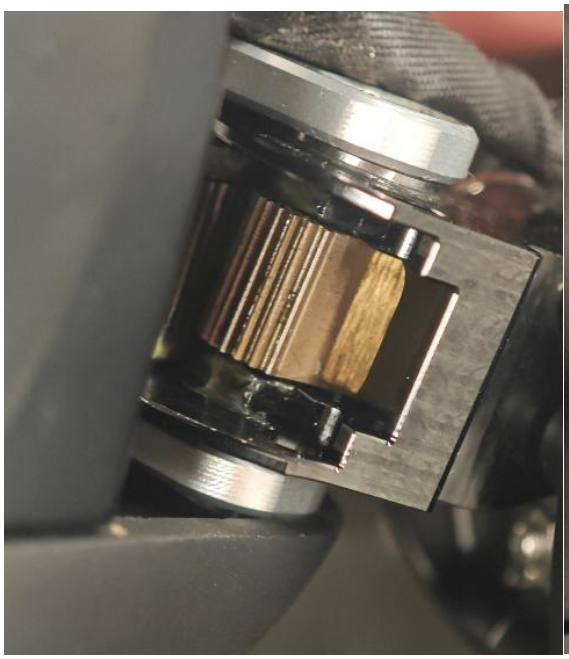
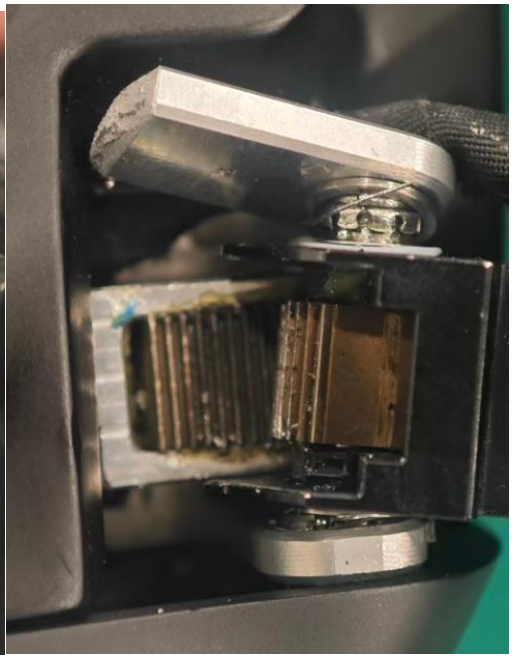
RH56DFTP Thumb

The thumb linkage is a wear-prone component easily damaged by heavy impact or drops. Severe linkage deformation or fracture requires factory return for repair.

RH56E2 Thumb

The E2 model features reinforced thumb linkage hardware but remains vulnerable to catastrophic impact damage. Significant structural deformation requires factory return service.

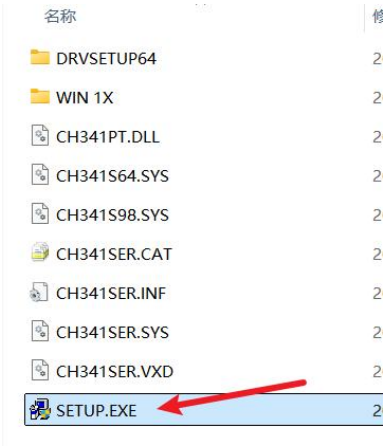
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III. Supplementary Information

Driver Installation

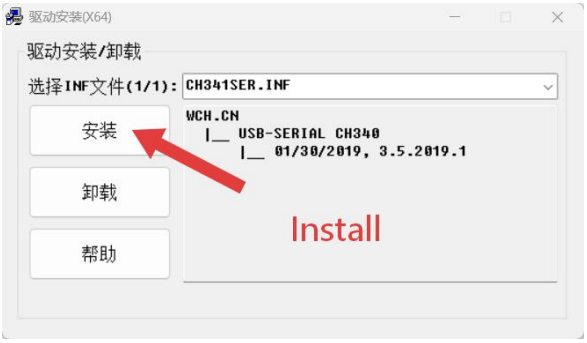
SETUP.EXEDownload the official CH340/CH341 driver package from the manufacturer or request it from technical support.

Extract the compressed file and run SETUP.EXE.

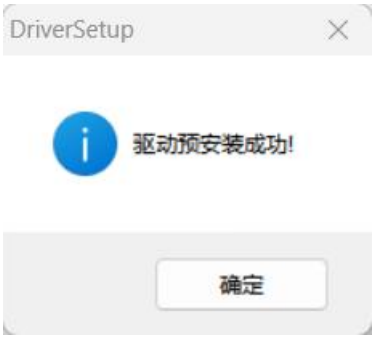


Allow the program to modify system device settings.

On the driver installer window, click Install.



A pop-up confirms successful pre-installation. After connecting a USB-to-serial adapter, the corresponding COM port will appear in Windows Device Manager.



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